Environ. Sci. Technol. **2008,** *42,* 153–158

Do Potential Methylation Rates Reflect Accumulated Methyl Mercury in Contaminated Sediments?

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Received June 28, 2007. Revised manuscript received September 24, 2007. Accepted September 27, 2007.

Relationships between the short-term mono-methyl mercury (MeHg) production, determined as the specific, potential methylation rate constant K_m (day^{-1}) after 48 h of incubation with isotope-enriched $^{201}Hg(II)$ at 23 °C, and the long-term accumulation of ambient MeHg, were investigated in contaminated sediments. The sediments covered a range of environments from small freshwater lakes to large brackish water estuaries and differed with respect to source and concentration of Hg, salinity, primary productivity, quantity and quality of organic matter, and temperature climate. Significant ($p < 0.001$), positive relationships were observed between K_m (day^{-1}) and the concentration of MeHg normalized to total Hg (%MeHg) for surface sediments (0–10, 0–15, and in one case 0–20 cm) across all environments, and across subsets of organic and minerogenic freshwaters. This suggests that the methylation process (MeHg production) overruled demethylation and net transport processes in the surface sediments. The lack of a relationship between K_m and %MeHg in two brackish water sediment depth profiles (0–100 cm) indicates that demethylation and the net effect of input−output are relatively more important at greater depths. Differences in the primary production and subsequent availability of easily degradable organic matter (serving as electron donor for methylating bacteria) was indicated to be the most important factor behind observed differences in %MeHg and K_m among sites. In contrast, concentrations of sulfate were not correlated to K_m, %MeHg, or absolute concentrations of MeHg. We conclude that total concentrations of Hg are of importance for the long-term accumulation of MeHg, and that %MeHg in surface sediments can be used as a proxy for the rate of methylation, across a range of sites from different environments.

Introduction

High concentrations of mono-methyl mercury (MeHg) in fish is a severe environmental problem (e.g., ref *1*). It is therefore of critical importance to understand and quantify the factors that control the concentration of MeHg in sediments, and the transfer of MeHg into the aquatic food web. The quantitative importance of methylation/demethylation processes, relative to input/output of MeHg, for the buildup of MeHg in sediments, is not fully understood and may differ among environments as well as among seasons.

The current view is that both methylation and demethylation primarily are biotic processes (*2*). While sulfate-reducing bacteria (SRB) have been identified as the primary methylators (*3–5*) in sediments, demethylation can be carried out by numerous types of microorganisms (e.g., *refs 6–8)*. Advances in analytical methodologies have made it possible to measure potential Hg methylation and MeHg demethylation rates using enriched stable isotope tracers of Hg(II) and MeHg (e.g., *refs 9 and 10)*. In studies of freshwater (*2*), estuarine (*11, 12*), and marine (*13*) surface sediments, a positive relationship has been observed between the potential methylation rate (expressed either as the rate, e.g., mol g^{-1} h^{-1}, or as the specific potential methylation rate constant, K_m, day^{-1}) and the concentration of MeHg (or %MeHg of total Hg). In near-shore marine surface sediments (*14*), no significant relationships were found. To our knowledge, the relationship between methylation and concentration of MeHg has not been evaluated in highly Hg-contaminated sediments, where remediation measures are being considered, nor has this relationship been evaluated across a range of sites from different environments (*2*).

The relationship between total concentrations of Hg and MeHg in sediments differs among studied systems. While it has been argued that factors other than total Hg are more important in controlling sediment MeHg concentration (*2, 15*), several studies have reported a significant, positive relationship between total Hg and MeHg (e.g., refs *13*, *14*, *16*, and *17*).

In this paper, we examine relationships between the specific, potential methylation rate constant (K_m, day^{-1}) and the absolute and relative concentrations of MeHg in contaminated sediments and evaluate the influence of factors known to influence the methylation process. Data from highly contaminated sediments (total Hg 1.0–1143 nmol g^{-1}) at eight contrasting sites, representing differences in source of Hg, salinity, primary productivity, organic matter content and quality, and temperature climate, are evaluated.

Material and Methods

Site Descriptions. Sediment samples were collected at eight sites in Sweden, during the period from October 2004 to October 2006 (Table S1, Supporting Information). All sites had been subjected to heavy, local Hg contamination. Four sites were contaminated by $Hg^0(l)$, mainly from the chlor-alkali industry, Köpmanholmen (Köp), Skutskär (Sku), Marnästjärn (Mar), and Ala-Lombolo (Ala), and four sites were contaminated by phenyl-Hg from the pulp and paper industry, Karlshäll (Kar), Turingen (Tur), Övre Svartsjön (Sva), and Nötöfjärden (Nöt). All sites except Mar and Ala had been subjected to pulp fiber discharge. The total Hg ranged between 1.0 and 1143 nmol g^{-1} (dry weight, dw) and reached the highest concentrations at sites with $Hg^0(l)$ as the source of contamination. In addition to sources of Hg, the sites represented gradients in salinity, annual air temperature sum, C/N ratio, and organic C content (Table S1, Supporting Information). They also represented a gradient ranging from small lakes (Mar, Tur, Sva, and Ala), via smaller freshwater-dominated estuaries (Kar and Nöt) to larger brackish water estuaries (Köp and Sku). As an estimate of the primary productivity at a certain site, we used the annual air

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10.1021/es0715851 CCC: $40.75 © 2008 American Chemical Society
Published on Web 11/30/2007

temperature sum in combination with the sediment C/N ratio. On the basis of productivity and salinity, the sites were divided into three data sets: (1) northern, low-productive freshwaters (Ala and Kar), (2) brackish waters with intermediate productivity (Köp and Sku), and (3) southern, high-productive freshwaters (Mar, Tur, Sva, and Nöt). There was a difference in sediment firmness among the sites, related to the composition of pulp fiber, natural organic matter, and minerogenic matter. On the basis of sediment firmness, pulp fiber content, and salinity, another division was (A) loose, organic, pulp-fiber-dominated freshwater sediments (Kar, Sva and Nöt); (B) firm, minerogenic freshwater sediments (Mar, Tur, and Ala); and (C) firm, pulp-fiber-enriched brackish water sediments (Köp and Sku).

Sampling, Onsite Measurements, and Sample Pretreatment. Polycarbonate core samplers were used for sediment sample collection, and at each sampling point, sediment cores were taken within a 1 m × 1 m area, were sectioned by depth and were pooled in 3–5 L plastic buckets. The pooled sediment samples were homogenized in the laboratory. After homogenization, pore water was extracted by centrifugation followed by filtration (0.45 μM) of the supernatant. Subsamples were taken for determination of methylation and demethylation rates and total concentrations. Oxidation of sediments during both sampling and pretreatment was minimized by working under a N_2(g) atmosphere. pH and dissolved H_2S concentrations were measured on site, as well as in the laboratory. A more thorough description of sampling and sample treatment is given in refs *18* and *19*.

Chemical Analyses of Pore Water and Sediment. Concentrations of MeHg and total Hg in pore water were determined by isotope dilution analysis as described by Fassett and Paulsen (*20*). MeHg was determined by an enriched (97.7%) $Me^{202}Hg$ standard (*21*) as described in refs *22* and *23*. Total Hg was determined using a ^{201}Hg(II)-enriched (98.11%) standard using U.S. EPA method 1631 (*24*). Method details for the pore water analysis of MeHg and total Hg and its quality assurance; methods for the determination of DOC, DIC, Cl, Br, and total S in pore water; and methods for the determination of total Hg, C, and N in sediments are described in ref *19*.

Determination of Potential Methylation and Demethylation Rates and Ambient MeHg Concentrations. Potential Hg methylation and MeHg demethylation rates were determined from a single analysis of the same sample by spiking with aqueous solutions of isotope-enriched $^{201}Hg(NO_3)_2$ (98.11%) and $Me^{204}HgCl$ (98.11%) to 40% and 0.2% of total Hg, respectively. For accurate determination of the (spiked) $Me^{204}Hg$ concentration at the start of the incubation period, all (spiked) samples were divided into two parts, of which one was immediately frozen at −20 °C, representing $t = 0$ days (t_0). The other part was incubated in darkness at 23 °C for 48 h in a glovebox under N_2(g), representing $t = 2$ days (t_2). The incubation time was based on a linear response of produced $Me^{201}Hg$ as a function of time. After 48 h, the incubation was stopped by freezing at −20 °C.

Prior to sample preparation, thawed sediment samples were spiked with an aqueous solution of $Me^{200}Hg$ to 0.5% of the total Hg concentration as a species-specific isotope standard for isotope dilution calibration. In addition, ^{199}Hg(II) was added to the samples at 40% of the total Hg concentration to correct for possible MeHg formation during sample treatment and analysis.

On the basis of the determined ambient MeHg concentration, the concentration of formed $Me^{201}Hg$ in the t_2 samples (methylated from the ^{201}Hg(II) spike during the incubation) was calculated from the measured 202/201 MeHg isotope ratio by reverse isotope dilution calculation. Correspondingly, demethylation of the added $Me^{204}Hg$ during incubation was calculated as the difference between the t_0 and t_2 $Me^{204}Hg$ concentrations, derived from the measured 202/204 MeHg isotope ratio, by reverse isotope dilution calculation. For details about the extraction and analysis procedures for MeHg, as well as the quality assurance of MeHg determinations, we refer to ref *19*.

First-order reactions were used to describe the potential methylation and demethylation rates (*10, 25*). Assuming that the contribution from demethylation of newly produced $Me^{201}Hg$ [from the ^{201}Hg(II) spike] is negligible within 48 h of spike addition (due to a very low initial $Me^{201}Hg$ concentration), the specific methylation rate constant (K_m) was calculated from the first-order equation: $K_m = [Me^{201}Hg]/([^{201}Hg]t)$, where $t = 2$ days. Similarly, assuming that methylation of demethylated ^{204}Hg(II) is very low during the 48 h incubation, the specific demethylation rate constant (K_d) was calculated from the first-order equation: $K_d = -\ln([Me^{204}Hg_{t_2}]/[Me^{204}Hg_{t_0}])/t$, where $[Me^{204}Hg_{t_2}]$ is the concentration of $Me^{204}Hg$ at the end of the incubation, $[Me^{204}Hg_{t_0}]$ is the concentration of $Me^{204}Hg$ at the start of the incubation, and $t = 2$ days.

Results and Discussion

Relationship between Total Hg and Total MeHg. The relationship between total concentrations of Hg and MeHg in sediment was significantly ($p = 0.001$, $n = 92$) positive for a data set comprising all sites, Figure 1a. Note, however, that the significant relationship was largely caused by a few data points. For the sites subjected to the most extensive sampling, Köp and Sku, significant relationships were observed for subsets of both surface sediments (0–10 cm at Köp, Figure 1b, $p < 0.001$) and depth profiles (0–40 cm at Köp and 0–100 cm at Sku, Figure 1c, $p < 0.001$; Figure 1d, $p = 0.001$). A significant relationship was also observed for surface sediments at 0–20 cm at site Kar (Figure 1e, $p = 0.001$). For the other sites (Sva, Nöt, Tur, Mar, and Ala), the relationship between total concentrations of Hg and MeHg was positive, but not significant ($p > 0.05$, data not shown). This was most probably due to lower absolute concentrations, and less variability in total Hg, in combination with few ($n = 4$–9) data points at each site.

The significant relationships shown in Figure 1a−e indicate that total Hg is one factor contributing to the production of MeHg in contaminated sediments. This observation is in agreement with previous reports (*2, 13, 14, 16, 17*), both from sediments contaminated by a local source and sediments affected by more diffuse Hg sources, and may be explained by a link between total concentration and Hg availability to methylating bacteria. This may appear to be in opposition to recent findings that concentrations of dissolved, neutral mercury sulfide [$Hg(SH)_2^0 + HgS^0$] species (*19*), and not total Hg in sediment pore water, control short-term methylation rates, as well as total MeHg concentrations in sediments. It should however be noted that, in the long term, and especially in data sets with large variations in Hg, there is an inevitable link between concentrations of total Hg and its bioavailable forms. This is best understood if the solubility of Hg is regulated by equilibria involving surface complexation reactions (with organic thiols or FeS(s) surfaces), generating a positive relationship between dissolved Hg (including bioavailable forms) and total sediment Hg. On the other hand, if HgS(s) or mixed Hg/FeS(s) phases control Hg solubility, the concentration of bioavailable, neutral mercury sulfides in pore water will be independent of total Hg concentration if dissolved inorganic sulfides are below 1 mM (*26*). Thus, in sediments with solid HgS(s) phases, absolute concentrations of Hg can be expected to have less of an impact on MeHg production.

Potential Rates of Methylation and Demethylation. With the exception of site Sku, the average specific, potential demethylation rate constant (K_d) was of a similar magnitude

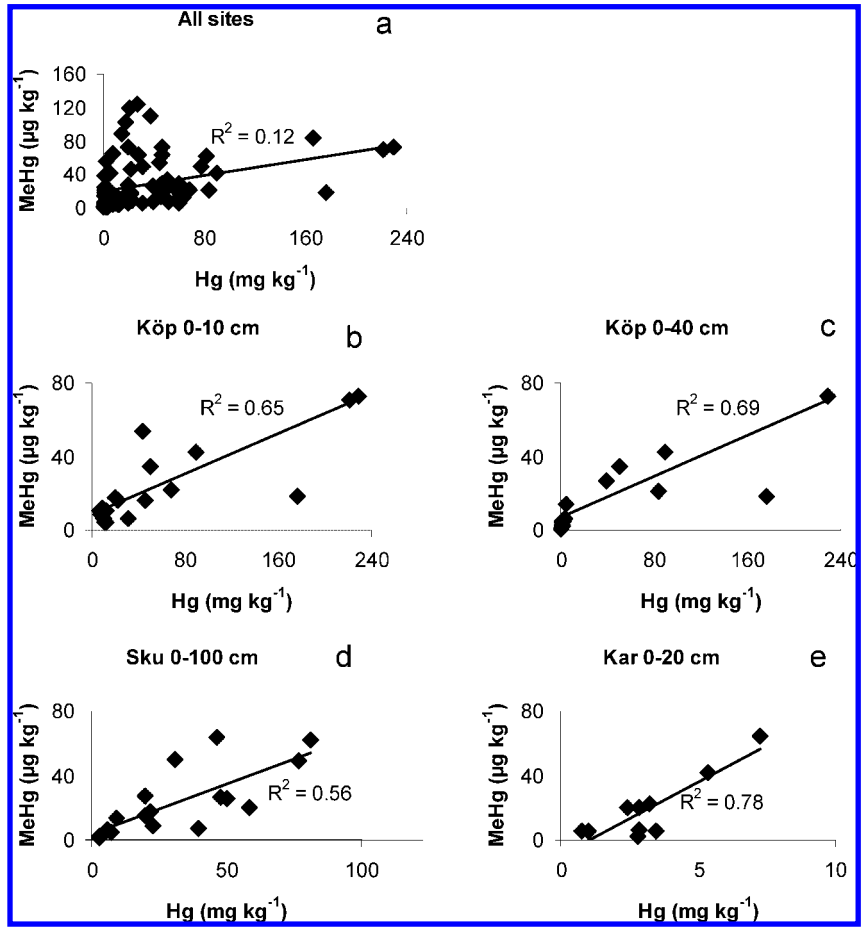


FIGURE 1. Relationship between total Hg and total MeHg for (a) all sites (n = 92), (b) Köp 0–10 cm (n = 16), (c) Köp 0–40 cm (n = 16), (d) Sku 0–100 cm (n = 17), and (e) Kar 0–20 cm (n = 10).

TABLE 1. Average Rate Constants (±SD) for Potential Methylation (K_m) and Potential Demethylation (K_d)

site	depth (cm)	K_m (day^{-1})	K_d (day^{-1})
Köp	0–10	0.002 ± 0.003[a]	0.07 ± 0.08[b]
	10–40	0.0002 ± 0.0001[c]	nd
Sku	0–20	0.004 ± 0.006[b]	0.54 ± 0.31[d]
	20–100	0.0004 ± 0.00004[e]	nd
Kar[d]	0–20	0.0004 ± 0.0006[f]	0.05 ± 0.06[e]
Ala[e]	0–15	0.002 ± 0.002[g]	0.02 ± 0.006[g]
all southern freshwaters[f]	0–10	0.02 ± 0.02[h]	0.12 ± 0.06[h]

[a] n = 22. [b] n = 16. [c] n = 10. [d] n = 5. [e] n = 6. [f] n = 10. [g] n = 9. [h] n = 19. nd = not determined.

at all sites. In contrast, average specific, potential methylation rate constants (K_m) varied by 2 orders of magnitude among sites (Table 1).

It should be noted that the values of K_d and K_m cannot be directly compared. The spike of ^{201}Hg added in the methylation experiment (40% of total Hg) is several orders of magnitude greater than the Me^{204}Hg spike (0.2% of total Hg) added in the demethylation experiment. Furthermore, methylation and demethylation proceed through very different mechanisms; for example, in the case of methylation, a passive uptake of neutral dissolved Hg(II) species has been suggested (*2, 19, 26*), while in the case of *mer*-induced demethylation, Hg is actively taken up by demethylating microbes (*27, 28*). Thus, the fraction of the added Hg and MeHg spikes available during the 48 h of methylation and demethylation, respectively, likely is very different. This restricts the possibilities, at this point, to determine true rates of methylation and demethylation, and subsequent net methylation rates (e.g., ref *10*). What can be done, however, is to compare K_d and K_m, respectively, determined in a similar way, among different sites.

Do Differences in Potential Methylation Rates Reflect Differences in MeHg Concentrations Among Sites? If variations in the rate of methylation (MeHg gross production) overrule variations in the sum of rates of demethylation and the net input−output of MeHg to the sediment, a positive relationship between the potential methylation rate constant (K_m) and the concentration of MeHg normalized to total Hg (%MeHg; 100 × [MeHg]/[Hg$_{tot}$]) in the sediment is expected. Note that the unit of K_m is expressed as the fraction of the Hg spike that is transformed to MeHg during 24 h at 23 °C. Thus, the unit of K_m is analogous to %MeHg, and K_m can be interpreted as reflecting the short-term MeHg production, while %MeHg can be interpreted as reflecting the long-term MeHg production. Additionally, a normalization of sediment MeHg to total Hg is reasonable given the significant correlation between MeHg and total Hg (Figure 1). Similarly, Hammerschmidt and Fitzgerald argued for a normalization of MeHg for variations in total Hg and reported a significant positive relationship between K_m and the ratio of MeHg to Hg(II) in marine surface sediments (*13*).

We report a positive, significant relationship between K_m and %MeHg for a data set comprising all surface sediments (0–20 cm) in this study ($p < 0.001$, Figure 2a). In contrast, no significant relationship was obtained between K_m and absolute MeHg concentrations ($R^2 = 0.02$, $n = 59$, $p = 0.21$, data not shown) for this data set, supporting the idea that MeHg concentrations need to be normalized for variations

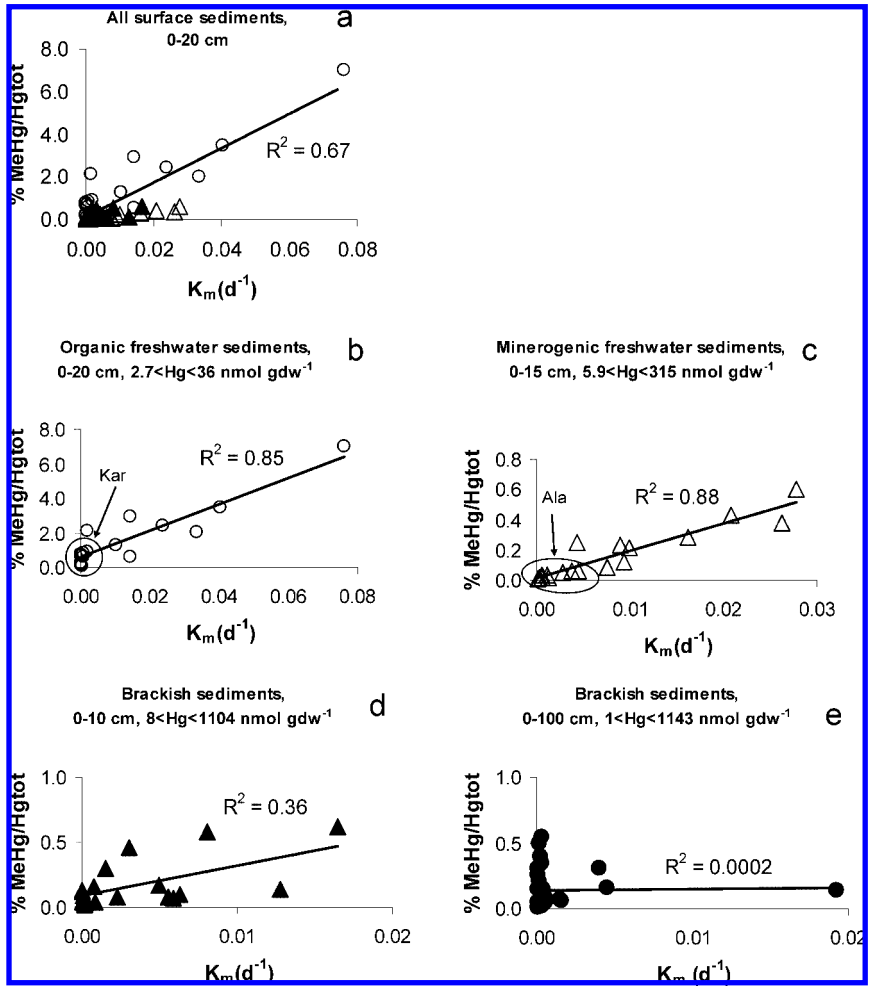


FIGURE 2. Relationship between K_m and percent MeHg of total Hg for (a) all surface sediments (open circles = organic freshwater sediments, open triangles = minerogenic freshwater sediments, filled triangles = brackish sediments, $n = 59$); (b) loose, organic, pulp-fiber-dominated freshwater sediments (Sva + Nöt + Kar, $n = 19$); (c) firm minerogenic freshwater sediments (Mar + Tur + Ala, $n = 19$); (d) firm, pulp-fiber-enriched brackish water surface sediments (Köp + Sku, $n = 21$); and (e) firm, pulp-fiber-enriched brackish water sediment depth profiles (Köp + Sku, $n = 33$).

in total Hg. When data were divided into different subsets, significant, positive relationships were obtained between K_m and %MeHg for loose, organic, pulp-fiber-dominated freshwater sediments ($p < 0.001$, Figure 2b); for firm, minerogenic freshwater sediments ($p < 0.001$, Figure 2c); and for firm, pulp-fiber-enriched brackish water surface sediments ($p = 0.006$, Figure 2d). In contrast, the relationship between K_m and total MeHg concentration was only significant for the firm, minerogenic freshwater sediments ($p = 0.004$, data not shown), but not for the loose organic freshwater sediments ($p = 0.43$) or the brackish water surface sediments ($p = 0.48$). For brackish water sediment samples extending to a depth of 100 cm, no significant relationship was observed between K_m and %MeHg ($p = 0.95$, Figure 2e).

The relationships in Figure 2 suggest that short-term potential methylation rate constants determined under laboratory conditions reflect a normalized long-term production of MeHg. In practice, this relationship implies that %MeHg (of total Hg) can be used as an estimate of long-term MeHg production in contaminated surface sediments. This is in line with the conclusion of Sunderland and co-workers (*11*), that %MeHg can be used to approximate the rate of methylation in estuarine sediments. In other studies (e.g., *refs 2 and 12)* weaker relationships between potential methylation rates and sediment MeHg have been reported. It should be noted that, in these studies, concentrations of MeHg were not normalized for differences in total Hg concentrations.

The lack of a relationship between K_m and %MeHg in brackish sediment depth profiles (Figure 2e) indicates that the methylation process is less dominant at greater sediment depths. As shown in Table 1, average K_m also decreases with depth by about 1 order of magnitude at the brackish water sites. The decrease in methylation with depth is in agreement with previous studies (e.g., *refs 9, 15, and 19)* and is explained by variations in the activity of SRB, which is often highest at the oxic/anoxic interface in sediments, where SRB are able to compete with respect to electron acceptors [e.g., O_2, Fe(III)] and have access to energy-rich organic matter (electron donor). Our finding that the relationship in the brackish surface sediments (Figure 2d) was also weaker, as compared to the freshwater sites, may be explained by a greater contribution from demethylation reactions at site Sku (Table 1).

Factors Influencing K_m and Accumulated MeHg in Contaminated Sediments. A comparison of the axes of Figure 2b–d reveals that %MeHg was about 1 order of magnitude greater, and K_m 2–5 times greater, in the organic freshwater sediments, as compared to the minerogenic freshwater and the brackish water sediments. The greater %MeHg may in part be explained by the fact that all organic freshwater sites had lower total concentrations of Hg. However, other factors,

known to influence the methylation process, may also contribute to the observed differences in %MeHg and K_m among sites.

In Figure 2b and c, data points from the two northernmost sites (Kar and Ala) are encircled. Both are freshwater sites, and Kar is truly oligotrophic. Both sites are also situated in a cold climate, as reflected by low annual air temperature sums (Kar at 942 °C and Ala at 522 °C, Table S1, Supporting Information). The cold climate, and in addition at Kar nutrient-poor conditions, will result in a low primary production, as compared to the more southerly situated sites. Thus, the low K_m and %MeHg at sites Kar and Ala can be related to a lesser availability of organic e^- donors to methylating bacteria. SRB use mainly low-molecular-weight fatty acids (e.g., acetate, butyrate, propionate, and lactate), alcohols, and H_2 as e^- donors and live in close connection with fermenters. An increased input of energy-rich organic matter to the sediment likely will increase the population size of the entire microbial community, including SRB. Besides the effect of temperature on primary production, it should also be noted that there is a direct decrease in the rate of all microbial processes as a result of lower temperatures, which will have an effect on %MeHg, but not on K_m, as the incubation was done at the same temperature (23 °C) for all sites. In addition to the differences in temperature climate, there is a difference in C/N ratio, with the average C/N ratio being greater at Kar (49) than at the two southernmost freshwater sites Sva and Nöt (16 and 33, respectively, Figure 2b and Table S1, Supporting Information). We have previously used the C/N ratio to indicate differences in primary productivity and energy-rich organic matter input among sites (*19*), with a low C/N ratio indicating a higher input of pelagic plankton and algae (with a C/N ratio around 5–10) to the sediment. Thus, it is likely that the difference in %MeHg between the oligotrophic estuary site Kar and the more eutrophic lakes Sva and Nöt is an effect of differences in input of energy-rich organic matter, which is affected both by temperature climate and by nutrient status. The C/N ratios at the three minerogenic freshwater sites (Ala, Tur, and Mar) were quite similar (Table S1, Supporting Information), indicating similar organic matter quality in the sediment; thus, the difference between these sites likely is mainly an effect of temperature climate.

Similarly, lower %MeHg at the brackish water estuarine sites Köp and Sku (Figure 2d), as compared to the smaller, eutrophic freshwater lakes Sva and Nöt (Figure 2b), can be explained by a lesser input of energy-rich organic matter to the brackish sediments. The high C/N ratio at Köp (186 on average, Table S1, Supporting Information) indicates that the sediment is highly dominated by pulp fiber, a poor substrate as an energy source for microorganisms. In contrast, the lower C/N ratio at the freshwater sites Mar, Sva, and Tur indicates a yearly input of pelagic plankton and algae, energy-rich organic matter that can serve as e^- donor and drive the microbial community. The site Tur is an exception in that it is southerly situated, mesotrophic, and has a low total Hg concentration, but still, the %MeHg is quite low. Likely, this is explained by a relatively low concentration of neutral mercury sulfides in the sediment pore water, as shown in a previous study (*19*).

In Figure 3, data points from site Ala, encircled in Figure 2c, are highlighted, showing a significant, positive relationship between K_m and %MeHg for this site alone. A more detailed evaluation reveals that both K_m and %MeHg decrease with depth in the sediment, a pattern previously described for the brackish sites as well as in earlier studies, which is mainly explained by deposition of energy-rich organic matter at the surface of the sediment and depletion further down. At 5–10 and 10–15 cm, variation in e^- donor availability with depth is less pronounced (Figure 3), and other factors, most importantly the concentration of neutral mercury sulfides, become important. We have shown in a previous study (*19*) that the concentration of neutral mercury sulfides explains a significant amount of the variation in K_m, especially below the top 5 cm, in sediments.

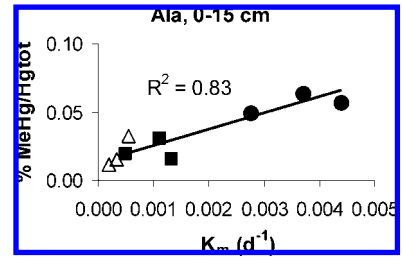


FIGURE 3. Relationship between K_m and %MeHg for 0–5 cm (filled circles), 5–10 cm (filled squares), and 10–15 cm (open triangles) at Ala (these data points are encircled in Figure 2c).

Our findings that short-term methylation rates, as determined in 48 h laboratory experiments, highly reflect the long-term buildup of MeHg in contaminated surface sediments across a range of sites with different properties (if expressed as %MeHg, i.e., normalized to total Hg), and that K_m is more variable than K_d among sites, suggest that rates of demethylation and input−output processes have less influence on MeHg accumulation than rates of methylation in these types of sediments. This also implies that relatively less complicated measurements of total MeHg and Hg in sediments may give very valuable information on MeHg production. The fact that variations in the specific potential methylation rate constant (K_m) in turn can be linked to factors operating at the level of bacterial metabolism (mainly e^- donor availability and bioavailability of mercury sulfides) further justifies the scientific basis for using %MeHg as a measure in risk assessments. It should be noted that the availability of sulfate (as e^- acceptor for SRB), which commonly has been suggested to limit MeHg production in sediments (e.g., *29 and 30)*, showed no significant ($p > 0.05$, Table S2, Figure S1, Supporting Information) relationships with either %MeHg, absolute concentrations of MeHg, or K_m in the sediments studied.

Acknowledgments

Bengt Andersson, Birgitta Olsson, Tom Larsson, and Torbjörn Karlsson are greatly acknowledged for assistance in the laboratory. This work was financed by the Swedish Environmental Protection Agency (contract No. 802-38-04F), North Sweden Soil Remediation Centre (MCN)−European Union Structural Funds and New Objective 1 (contract No. 113-12534-00), and the Centre for Environmental Research (CMF) in Umeå.

Supporting Information Available

A table with selected chemical parameters, ranges in concentrations of sulfate, and a figure with sulfate versus MeHg for all sites are presented as Supporting Information. This material is available free of charge via the Internet at http://pubs.acs.org.

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ES0715851